6G Weekly Survey — Jun 15 - Jun 21, 2026

24 items across 6 topics, ranked by importance.

AI-native air interface

Game-Theoretic Latent Space Alignment for Multi-user Semantic MIMO Communications

Giuseppe Di Poce, Mattia Merluzzi, Emilio Calvanese Strinati, Paolo Di Lorenzo · arXiv cs.IT · 2026-06-10

★★★★■ Addresses a fundamental problem in semantic communications—heterogeneous latent spaces—with rigorous theoretical analysis and practical cognitive radio integration, highly relevant to AI-native 6G air interfaces where diverse AI models must coexist.

This paper addresses semantic mismatch in multi-user MIMO interference networks where independently trained AI agents use heterogeneous latent representations. The authors formulate semantic alignment as a non-cooperative game and derive a closed-form solution for joint optimization of linear semantic MIMO transceivers under cognitive radio constraints (primary/secondary user hierarchy). They develop a semantic water-filling algorithm with provable convergence to Nash equilibrium, linking semantic alignment properties to physical-layer interference management.

Contribution: First game-theoretic framework combining semantic latent-space alignment with physical-layer interference mitigation in multi-user MIMO, providing closed-form solutions and convergence guarantees for the joint semantic-physical optimization problem.

Relevance to your work: Directly addresses semantic/task-oriented communication in your research portfolio, showing how to reconcile independently trained neural transceivers in multi-user MIMO scenarios—critical for practical AI-native PHY deployment where semantic interoperability cannot be assumed.

Orchestrating the Twin Transition in Multinational Corporations: Technology Roadmapping for Green and Digital Global Business Services

Han-Teng Liao, Karen Ang · arXiv eess.SY · 2026-06-11

★■■■■ Business-process and policy-focused work with no technical contributions to wireless systems, PHY layer design, or communications engineering.

This paper examines how multinational corporations orchestrate green and digital transformation in Global Business Services units by synthesizing Technology Roadmapping with ITU innovation frameworks. A bibliometric analysis traces evolution from basic automation toward 'Sustainable Intelligence' and maps stakeholder engagement across geopolitical contexts. The work proposes GBS as an 'operational airlock' mediating landscape pressures (EU mandates, carbon mechanisms) and AI-native workflow innovations.

Contribution: Proposes a socio-technical framework linking Technology Roadmapping to ITU toolkits for coordinating dual green-digital transitions in multinational service operations.

Relevance to your work: tangential

6G Demos & Trials

Quantization Limitations of Leakage Suppression in Self-Calibrating Monostatic Integrated Sensing and Communication MIMO Systems

Jan Adler, Florian Gast, Gerhard P. Fettweis, Rafael F. Schaefer · arXiv eess.SP · 2026-06-10

★★★★■ Addresses a critical deployment barrier for monostatic ISAC (essential for 6G joint sensing-communication) with hardware-validated theory from a credible German research group.

This paper addresses transmit-to-receive leakage in monostatic ISAC MIMO systems, where digital precoding theoretically suppresses self-interference but fails in practice. The authors derive a closed-form model showing quantization noise in the RF chain as the primary limitation, then validate predictions against hardware measurements. Results demonstrate fundamental performance bounds for digitally-precoded leakage cancellation.

Contribution: First analytical characterization of how DAC/ADC quantization noise fundamentally limits digital precoding-based self-interference cancellation in practical monostatic ISAC systems.

Relevance to your work: Directly relevant to your ISAC/JCAS interests—exposes fundamental hardware constraints on full-duplex monostatic sensing that affect system-level design trade-offs between sensing dynamic range and communication performance.

LLM-Enabled NWDAF: A Step Toward AI-Native 6G Network Intelligence

Henok Daniel, Omar Alhussein, Cheng Li, Jie Liang, et al. · arXiv cs.NI · 2026-06-10

★★★■■ Solid contribution to open-source 6G infrastructure and demonstrates practical LLM integration for network operations, though limited to intent classification over seven categories rather than advancing core AI-native air interface techniques.

This work presents an open-source NWDAF implementation for Free5GC that integrates an LLM-based conversational interface to enable natural language control of network analytics and event subscriptions. The system uses semantic embedding to map user intents to seven predefined categories, triggering analytics queries or subscription commands to AMF/SMF network functions via Prometheus monitoring. The approach aims to simplify operator interaction with 5G/6G core network management by abstracting traditional interfaces through intent-driven AI.

Contribution: First open-source NWDAF with integrated LLM interface for intent-based network analytics control, demonstrating practical AI-native management for 5G/6G core networks.

Relevance to your work: Tangential—focuses on core network automation and AI-driven management rather than physical layer, channel modeling, or wireless air interface innovation central to your Beyond-5G research.

Sub-array Selection Optimization for Joint Self-Interference and Multi-User Interference Suppression in FD mMIMO

Yuanzhe Gong, Yuanxing Zhang, Tho Le-Ngoc · arXiv eess.SP · 2026-06-11

★★★■■ Solid experimental work on full-duplex mMIMO with measurable SI suppression gains, though full-duplex deployment remains uncertain for 6G and the PSO approach is incremental.

This paper addresses self-interference (SI) and multi-user interference (MUI) in full-duplex massive MIMO by proposing a joint sub-array selection and angular perturbation-based nulling scheme optimized via particle swarm optimization. An 8x8 Tx-Rx FD array prototype demonstrates that SI channels vary significantly across sub-arrays, enabling selective exploitation. Experimental results show 29-31 dB residual SI suppression improvements, achieving 83-85 dB average beam-level isolation and MUI suppression below -181 dB.

Contribution: The work empirically validates that sub-array selection exploits spatial variation in SI channels to enhance full-duplex mMIMO isolation by 30+ dB, complementing traditional beamforming nulling techniques.

Relevance to your work: Tangential—full-duplex is outside your core focus on AI-native PHY, ISAC, and NTN, though mMIMO beam management methods may offer transferable insights for spatial interference mitigation.

What Drives Test-Time Adaptation for CLIP? A Controlled Empirical Study from an Update Perspective

Jiazhen Huang, Xiao Chen, Zhiming Liu, Yaru Sun, et al. · arXiv cs.LG · 2026-06-12

★★■■■ While rigorous as a benchmark study for computer vision models, this work addresses test-time adaptation for image-language models with no direct connection to wireless physical layer, channel modeling, or air interface design.
This paper conducts a controlled empirical study of test-time adaptation (TTA) methods for CLIP vision-language models under distribution shift. The authors organize existing TTA4CLIP methods into three paradigms based on what is updated at test time, introduce TTABC benchmark integrating 20+ methods, and systematically analyze what drives adaptation performance. Key findings show adaptation gains come primarily from test-time evidence and reliable proxies rather than heavy optimization, and no single paradigm is universally optimal across different types of distribution shifts.

Contribution: First systematic empirical study providing a unified framework and open benchmark (TTABC) to understand what mechanisms actually drive test-time adaptation in vision-language models, revealing that lightweight evidence-based updates can match heavy optimization approaches.

Relevance to your work: tangential

Feasibility Assessment of Remote Driving via Latency Analysis of ITS-G5 and Cellular Networks in the MASA Living Lab

Gaetano Orazio Cauchi, Antonio Solida, Salvatore Iandolo, Marco Savarese, et al. · arXiv cs.NI · 2026-06-11

★★■■■ Competent empirical study of existing technologies (ITS-G5, 5G NSA/SA) for V2X teleoperation, but limited novelty in methodology or direct relevance to emerging 6G air-interface research.
Presents field measurements of ITS-G5 and 5G latency for remote vehicle teleoperation in the MASA city-scale testbed, focusing on uplink video and downlink control under varying loads. Characterizes latency distributions and reliability for both technologies across realistic coverage scenarios. Findings inform hybrid communication strategies combining dedicated short-range and cellular connectivity for safe remote driving.

Contribution: Real-world latency characterization of ITS-G5 vs. 5G in a deployed intelligent transportation infrastructure, revealing technology-specific performance trade-offs under operational network load.

Relevance to your work: Tangential—focuses on current standardized V2X protocols rather than AI-native PHY, sub-THz waveforms, or 6G architectural innovations your research targets.

Greenness-Driven Scheduling in Far Edge Kubernetes: A CODECO Evaluation

Kaikang Huang, Dalal Ali, Rute C. Sofia · arXiv cs.NI · 2026-06-10

★★■■■ Competent energy-aware orchestration work for edge computing, but addresses infrastructure-layer concerns orthogonal to air interface innovation and lacks direct applicability to wireless PHY or 6G system design.
This paper evaluates CODECO, a Kubernetes scheduler that optimizes container placement using cross-layer energy metrics (compute via Kepler + network/IP-level) to minimize energy consumption in IoT-Edge-Cloud deployments. Experimental validation on ARM-based far-edge devices shows up to 11 mJ computational and 4.14 mJ network energy savings versus vanilla Kubernetes under diverse fault conditions (CPU stress, asymmetric delay, bandwidth contention). A composite greenness score combining both energy dimensions provides consistent scheduling guidance across scenarios.

Contribution: The ILP-based scheduler jointly optimizes compute and network energy consumption using real-time cross-layer metrics, validated on realistic embedded edge hardware under fault injection.

Relevance to your work: Tangential—this is Kubernetes/edge-cloud orchestration research; it does not address AI-native air interfaces, channel modeling, ISAC, NTN, or any physical-layer aspects relevant to Beyond-5G wireless systems.

Patching Control Lyapunov Barrier Functions for Temporal Logic Specifications with Bounded Controls

Ruikun Zhou, Yating Yuan, Haocheng Chang, Yinan Li, et al. · arXiv eess.SY · 2026-06-11

★★■■■ This is solid control-theoretic work with formal verification guarantees, but its scope is narrow and primarily relevant to robotics path planning rather than wireless communications or 6G physical layer design.

This paper proposes a control synthesis framework for continuous-time systems under Linear Temporal Logic specifications and bounded control inputs, combining LTL task decomposition with Control Lyapunov-Barrier Functions. The method constructs switching feedback controllers that enable online planning without explicit state-space abstraction, ensuring formal verification and robust continuous specification satisfaction. Validation includes numerical simulations and hardware demonstration on a Crazyflie quadrotor.

Contribution: The novel contribution is an abstraction-free approach that patches CLBF level sets to satisfy decomposed LTL subtasks, enabling formally verified switching controllers with efficient online re-planning under state perturbations.

Relevance to your work: Tangential—while formal methods for network automation appear in 6G discussions, this robotics-focused control synthesis work does not directly address AI-native air interfaces, channel modeling, or wireless system design challenges central to your research.

6G Physical Layer

Towards Standardizing Affine Frequency Division Multiplexing (AFDM) for Future Wireless Networks

Qu Luo, Lixia Xiao, Pei Xiao, Zilong Liu, et al. · arXiv eess.SP · 2026-06-11

★★★★■ Strong contribution from credible research group addressing standardization feasibility of a leading 6G waveform candidate, with practical analysis of multi-antenna/multi-user scenarios and concrete compatibility demonstrations that directly inform 3GPP discussions.

This paper systematically evaluates AFDM waveform for 6G standardization, demonstrating backward compatibility with 4G/5G multi-numerology frameworks, FMCW radar, and LoRa while highlighting robustness to doubly-selective channels. The authors examine practical implementation aspects including MIMO/multi-user support and PAPR characteristics, then validate AFDM's suitability for NTN, ISAC, V2X, and underwater scenarios with severe delay-Doppler spread. The work positions AFDM as a practical candidate for 3GPP evolution beyond OFDM.

Contribution: First comprehensive standardization-oriented treatment of AFDM showing concrete backward compatibility paths with existing 3GPP numerologies and radar waveforms, addressing the critical gap between theoretical AFDM advantages and practical deployment.

Relevance to your work: Directly relevant to your 6G waveform and ISAC research interests, particularly the dual-functional communication-sensing capability and NTN deployment scenarios where AFDM's delay-Doppler resilience addresses key beyond-5G physical layer challenges.

Jamming-Resilient Sparse Delay-Doppler NOMA: Unitary Precoding, Randomized Active Sets, and Superincreasing Power Allocation

Michel Kulhandjian, Hovannes Kulhandjian, Theodoros A. Tsiftsis · arXiv eess.SP · 2026-06-08

★★★■■ Solid theoretical analysis of jamming-resilient OTFS-NOMA with proven robustness, but the military-oriented anti-jam focus and limited experimental validation limit immediate 6G standardization impact.

The paper proposes a sparse delay-Doppler NOMA scheme that randomly hops user data across delay-Doppler bins using shared pseudo-random seeds and unitary precoding, making it resilient to intentional jamming. A Marchenko-Pastur conditioning argument proves that random unitary submatrices remain well-conditioned even when the jammer knows the hopping pattern, avoiding the partial-band BER floor of conventional OTFS-NOMA. For multi-user scenarios, superincreasing power allocation enables low-complexity SIC that provably matches ML detection, achieving ~40 dB BER improvement against pattern-aware jammers.

Contribution: The use of Marchenko-Pastur random matrix theory to prove that jammer knowledge of the hopping seed does not break the system, combined with superincreasing power allocation that makes SIC equivalent to ML detection without error propagation.

Relevance to your work: Directly relevant to 6G waveform research (OTFS variants) and anti-jam PHY design for non-terrestrial networks where intentional interference is a concern, though the NOMA aspect is less central to current 6G consensus.

Space-Based GNSS Radio Frequency Interference Detection Evaluation Through Multi-Satellite Data Integration

Anouar Boumeftah, Peter Klimas, Gunes Karabulut Kurt · arXiv eess.SP · 2026-06-12

★★■■■ Competent analysis of RFI detection scaling, but focused on LEO-based GNSS-R monitoring rather than communication system design or physical-layer innovation.

This paper evaluates how constellation size affects radio frequency interference (RFI) detection performance in space-based GNSS reflectometry systems, analyzing three months of delay-Doppler observations from seven CYGNSS spacecraft. A seven-satellite constellation reduces median detection latency by 4.7× versus a single satellite, increases interception probability from 2% to 11.5%, and improves revisit time from 5.8 hours to under 2 hours. Analysis shows three satellites as the minimum effective constellation size, with diminishing returns beyond that.

Contribution: First empirical quantification of constellation-size scaling laws for space-based terrestrial RFI detection using multi-month CYGNSS data, establishing three satellites as the minimum viable deployment.

Relevance to your work: tangential

Core network & 6G core

Noncoherent ISAC over Block-Fading Channels: Asymptotic Performance Analysis

Hao Yang, Kai Wan, Giuseppe Caire · arXiv cs.IT · 2026-06-12

★★★★■ Strong theoretical contribution from Caire's group on fundamental ISAC limits in the practically relevant noncoherent regime, with rigorous asymptotic analysis revealing regime-dependent tradeoff behavior that informs system design.

This work analyzes the fundamental limits of joint sensing and communication (ISAC) over block-fading MIMO channels without channel state information at transmitter or receiver. In the high-SNR regime, the authors derive bounds on noncoherent mutual information and optimize spatial power allocation to balance communication (requiring unitary space-time modulation) and sensing objectives. At low SNR, asymptotic analysis reveals that rank-one transmission along the dominant target-response eigenvector is optimal for sensing while incurring no first-order communication penalty, eliminating the sensing-communication tradeoff.

Contribution: Proves that the fundamental sensing-communication tradeoff in ISAC disappears asymptotically at low SNR, where optimal sensing beamforming (rank-one along dominant eigenvector) causes zero first-order communication loss—a surprising synergy absent at high SNR.

Relevance to your work: Directly relevant to your ISAC research interests, providing fundamental characterization of sensing-communication tradeoffs under realistic assumptions (no CSI) and regime-specific design principles for spatial beamforming in joint radar-communication systems.

Rotatable Antenna-Enabled Near-Field Integrated Sensing and Communication

Zequan Wang, Liang Yin, Yitong Liu, Hongwen Yang · arXiv eess.SP · 2026-06-11

★★★■■ Solid incremental contribution introducing a practical hardware dimension (mechanical rotation) to near-field ISAC, though deployment complexity and mechanical reliability constraints limit immediate 6G standardization impact.

This paper introduces rotatable antennas (RAs) as a novel spatial degree-of-freedom for near-field ISAC systems with sub-connected hybrid beamforming, where each antenna element can independently rotate its boresight direction. An alternating optimization framework combining fractional programming, Riemannian optimization, and spherical-cap Frank-Wolfe methods jointly designs beamformers and antenna orientations while deriving closed-form Cramér-Rao bounds for sensing performance. Simulations show that element-wise rotation can compensate for limited RF chains, matching fully-digital performance in some regimes while improving off-broadside sensing accuracy and clutter suppression.

Contribution: First work to exploit element-wise antenna rotation as an orientation-domain spatial DoF in near-field ISAC, demonstrating that mechanical rotation can compensate for RF chain limitations in hybrid architectures.

Relevance to your work: Directly relevant to ISAC research, particularly for near-field beamforming and sensing-communication tradeoffs; the hybrid beamforming architecture and orientation-domain DoF could inform practical massive MIMO deployments where RF chain count is limited by cost.

Generalized Framework for a Fair Comparison of Cellular and Cooperative Massive MIMO Systems

Leonard Paul Schulz, Stefan Schwarz, Gerhard Bauch · arXiv cs.IT · 2026-06-12

★★★■■ Solid methodological contribution that clarifies architectural trade-offs in massive MIMO deployments, though primarily addresses known distributed/centralized system design principles rather than introducing fundamentally new techniques.

This paper presents a graph-based framework to fairly compare cellular, coordinated, and cell-free massive MIMO architectures by systematically separating antenna distribution from inter-site cooperation. The authors derive compatible uplink/downlink spectral efficiency expressions and demonstrate through large-scale simulations (≥2.5×2.5 km²) that coordinated phase-aligned beamforming across distributed antennas is the primary driver of cooperation gains. Results show that coordinated DAS and hybrid cell-free systems achieve near-optimal performance with significantly relaxed midhaul requirements compared to fully centralized cell-free processing.

Contribution: The framework isolates antenna distribution from cooperation topology, enabling rigorous comparison of seven massive MIMO architectures and revealing that coordination strategies—not just antenna placement—primarily determine performance gains.

Relevance to your work: Tangential—the work provides valuable simulation methodology and architectural insights for massive MIMO but does not directly address AI-native physical layers, generative channel models, or next-generation waveforms central to your Beyond-5G research focus.

Beamforming Design for Stem-Connected Microwave Linear Analog Computer (MiLAC)-Aided Multiuser MISO Downlinks

Yuchen Zhang, Zheyu Wu, Bruno Clerckx, Tareq Y. Al-Naffouri · arXiv eess.SP · 2026-06-12

★★★■■ Solid contribution to analog beamforming architectures with rigorous theoretical analysis and practical optimization, but represents incremental refinement of hybrid analog-digital methods rather than breakthrough AI-native PHY innovation.

The paper analyzes stem-connected Microwave Linear Analog Computers (MiLAC) for analog beamforming in multiuser MISO downlinks, showing they can realize any beamformer on the complex Stiefel manifold with linear rather than quadratic scaling in antenna count. A Riemannian-manifold-based WMMSE solver is developed to handle discrete, bounded hardware constraints. Simulations demonstrate that stem-connected MiLACs match fully-connected performance while achieving near-digital sum rates without symbol-rate digital processing.

Contribution: Proves that stem-connected MiLACs with $N \geq 2K-1$ achieve the same multiuser sum-rate as fully-connected architectures while requiring only linear complexity in antenna count, and provides a practical optimization framework for discrete hardware constraints.

Relevance to your work: Relevant as analog processing architectures may complement learned end-to-end receivers by reducing digital processing requirements in massive MIMO scenarios, though the fixed-structure analog network differs from fully adaptive neural transceivers.

Network automation & AIOps

STCC: A Unified Source-Channel Semantic Token Coding Framework for Semantic Communications

Zhicheng Bao, Chen Dong, Sen Wang, Long Liu, et al. · arXiv cs.IT · 2026-06-10

★★★★■ Strong contribution to semantic/task-oriented communication with a principled approach to bridging foundation models and physical layer, though practical deployment complexity and scaling to diverse semantic spaces remain open questions.

The paper proposes STCC, a joint source-channel coding framework that transmits discrete semantic tokens (from foundation models) over noisy channels using learned, geometrically structured constellations instead of fixed modulation schemes. The core STC codec employs an MLP-based encoder with triple-loss optimization to align channel topology with semantic embedding space, transforming channel noise into semantically similar token errors rather than random corruption. Experiments show significant gains over traditional systems at low SNR without requiring receiver modifications.

Contribution: Novel integration of discrete semantic token transmission with learned constellation design that geometrically structures the channel space to match semantic embeddings, ensuring graceful degradation under noise.

Relevance to your work: Directly relevant to your semantic communication interests and AI-native air interface design, particularly the learned PHY components and task-oriented paradigms for 6G semantic layers.

NARRAS: Edge-Triggered Distributed Inference for CSI-Based Localization in Vehicular IoT Networks

Rodrigo Oliver, Ricardo Vazquez Alvarez, Alejandro Lancho, Stefano Rini · arXiv eess.SP · 2026-06-10

★★★★■ Strong contribution connecting task-oriented communication, learned CSI compression, and distributed sensing with practical vehicular-IoT constraints; novel use of chart regularization for robust sparse reporting aligns well with 6G resource-efficiency goals.

The paper addresses CSI-based localization in vehicular IoT where distributed remote antenna arrays must selectively report channel observations to a fusion center over a shared uplink with limited capacity. NARRAS is a learned edge-triggered policy where each array uses recurrent summaries and memory of past transmissions to decide locally whether to report, trained with differentiable activity penalties and channel-chart regularization to enforce geometric structure in latent features. Experiments show NARRAS achieves better localization accuracy than heuristic and learned baselines at matched uplink activity budgets, with chart regularization particularly reducing tail errors in sparse-reporting regimes.

Contribution: First application of edge-triggered distributed inference with explicit activity budgeting and channel-chart regularization to CSI-based localization, demonstrating that geometry-aware learned reporting policies outperform dense and heuristic sparse strategies.

Relevance to your work: Directly relevant to semantic/task-oriented communication and learned CSI compression; the edge-triggered inference framework and latent geometry constraints offer techniques applicable to distributed ISAC and AI-native PHY resource management.

LGVSC: A Large-Model-Driven Generative Video Semantic Communication Framework

Yu Ma, Hang Yin, Li Qiao, Shuo Sun, et al. · arXiv eess.SP · 2026-06-11

★★★■■ Solid demonstration of semantic communication principles with large models, but practical deployment challenges (computational cost, latency, model complexity) and lack of comparison to neural compression baselines limit immediate impact on 6G standardization.

This paper presents LGVSC, a semantic communication framework for video transmission that uses multimodal large language models to extract keyframes at the transmitter and regenerate video at the receiver, achieving extremely low bandwidth usage (10^-4 to 10^-3 channel ratio). The framework introduces a probability-based semantic similarity score (PSSS) for evaluation and maintains interpretability through explicit intermediate semantic representations rather than end-to-end black-box learning. Simulations show strong performance and zero-shot generalization on downstream tasks while preserving semantic fidelity.

Contribution: Novel integration of multimodal large models for semantic-guided keyframe extraction and generative video reconstruction, achieving orders-of-magnitude bandwidth reduction with interpretable intermediate representations and a new continuous semantic similarity metric.

Relevance to your work: Directly relevant to your semantic and task-oriented communication interests, particularly the shift from bit-level to semantic-level transmission and the use of generative AI for wireless PHY, though focuses on application layer rather than physical layer AI-native design.

Designed-Source Reductions and a Dual-Purpose Feasibility Band for Semantic Rate-Distortion

Joss Armstrong · arXiv cs.IT · 2026-06-09

★★★■■ Solid theoretical refinement of semantic communication foundations that clarifies when deterministic task structures permit tighter analysis, though impact is limited to the niche of formal semantic rate-distortion theory rather than practical 6G PHY design.

This paper extends the Stavrou-Kountouris semantic rate-distortion framework from stochastic sources to 'designed sources' where the semantic object is a deterministic oracle allocation rather than a random variable. Under smooth concave utility and deterministic common-category encoders, the author shows the SK decoder specializes to conditional-mean decoding and derives a feasibility band for aggregate verification tasks that bounds the achievable rate by partition cardinality. The analysis clarifies when semantic communication can leverage deterministic task structures rather than treating semantics as stochastic.

Contribution: Identifies a subclass of task-oriented systems (designed sources with deterministic oracle allocations) where semantic rate-distortion theory simplifies to classical Lloyd-Max quantization, and derives a dual-purpose feasibility band when the second fidelity metric is aggregate verification rather than point-wise distortion.

Relevance to your work: Connects to semantic and task-oriented communication interests by formalizing the distinction between stochastic semantic sources and deterministic task allocations, which may inform how AI-native air interfaces handle goal-oriented transmission versus content delivery.

ISAC (sensing + communication)

Max-Min Secrecy Rate Optimization for Secure ISAC Networks: Global Optimization and Low-Complexity Algorithm

Thanh-Nha To, Trung Quang Pham, Dang Y Hoang, Hoang-Lai Pham, et al. · arXiv cs.IT · 2026-06-11

★★★■■ Solid theoretical contribution addressing security-sensing trade-offs in ISAC, but limited to beampattern matching metrics rather than emerging information-theoretic or task-oriented ISAC frameworks.

This paper addresses physical-layer security in multi-user ISAC systems where sensing users are untrusted potential eavesdroppers. The authors formulate a max-min secrecy rate problem with beampattern matching constraints for sensing quality, developing both a branch-and-bound global optimizer and a low-complexity successive convex approximation algorithm. Numerical results show the SCA method achieves near-optimal secrecy rates with substantially lower complexity than global optimization.

Contribution: First joint treatment of fairness-based secrecy rate maximization under beampattern constraints for ISAC, with globally optimal benchmark and practical SCA algorithm.

Relevance to your work: Directly relevant to ISAC research, particularly for understanding fundamental security-sensing trade-offs and practical beamforming design when sensing nodes are untrusted.

Inverse Learning assisted V2I Communication for Intent Based 6G ISAC Vehicular Networks

Anoop C, Anup Aprem · arXiv eess.SP · 2026-06-11

★★★■■ Addresses a relevant 6G ISAC use case with novel inverse learning methods, though limited to vehicular scenarios and lacks experimental validation on real ISAC hardware or standardized channel models.

This paper addresses V2I communication in 6G ISAC vehicular networks where roadside units (RSUs) act as autonomous agents with unknown intent-based utility functions. The authors propose inverse learning methods (nonparametric ATIL and parametric FICNNIL/PICNNIL) to infer RSU utility functions from observed ISAC actions like adaptive beamwidth allocation, plus federated variants. Demonstrations include predictive scheduling for cooperative data downloading and dynamic cluster-head selection in vehicular micro-clouds.

Contribution: First inverse learning framework to infer autonomous RSU decision-making utilities from ISAC observations, enabling vehicles to predict and adapt to intent-based 6G infrastructure behavior.

Relevance to your work: Directly relevant to ISAC research by demonstrating how sensing outputs (vehicle kinematics) inform communication decisions (beamwidth allocation), though the inverse learning approach is more network-layer focused than physical-layer ISAC design.

Deep Reinforcement Learning for Adaptive Power Allocation in ISAC Systems with Mobile Target

Zhilin Fu, Sangmin Kim, Sangwon Hwang, Jihwan Moon, et al. · arXiv eess.SP · 2026-06-10

★★★■■ Solid incremental contribution applying established DRL methods to ISAC resource allocation with mobile targets; well-motivated but limited novelty in the underlying RL approach and no experimental validation.

This work addresses dynamic power allocation in an ISAC system tracking a mobile target by formulating it as a Markov decision process solved with soft actor-critic (SAC) deep reinforcement learning. The authors employ a Dirichlet policy to generate normalized continuous power allocation actions and design a reward function that balances sensing (tracking) and communication objectives under resource constraints. Simulations show improved tracking performance versus baselines while maintaining communication quality.

Contribution: The novel combination of SAC-based DRL with a Dirichlet policy for continuous normalized power allocation in ISAC, coupled with a carefully crafted reward function that dynamically trades off sensing versus communication under mobile target uncertainty.

Relevance to your work: Directly relevant to your ISAC research interests, particularly for dynamic resource management in joint sensing-communication scenarios; the RL-based adaptive allocation framework could inform future AI-native ISAC air interface designs.

Coherent Multiband OFDM Sensing via Low-Complexity Gap Reconstruction

Lorenzo Pucci, Leonardo Pucci, Andrea Giorgetti · arXiv eess.SP · 2026-06-09

★★★■■ Addresses a practical ISAC spectrum-sharing problem with a scalable solution, though limited to specific dual-subband geometry and lacks experimental validation or integration with 6G waveform discussions.

The paper tackles coherent multiband OFDM sensing in ISAC systems where two sensing subbands flank a central communication band, creating spectral gaps that cause grating lobes in delay profiles. The authors propose a low-complexity iterative method combining delay-domain equalization with apodization-based reconstruction and data-consistency enforcement to fill spectral gaps. Results show performance approaching full-band sensing for moderate gaps, with better scaling than compressed-sensing baselines (OMP) in multi-target, low-SNR scenarios.

Contribution: A computationally efficient gap-reconstruction algorithm for dual-band OFDM sensing that avoids target-count-dependent complexity of compressed sensing while maintaining competitive performance in practical low-SNR, multi-target regimes.

Relevance to your work: Directly relevant to your ISAC research, particularly for OFDM-based joint communication-sensing architectures where spectral efficiency demands non-contiguous sensing bands and low-complexity processing for real-time target detection.